

Lab 1: Multiplier 4-bit

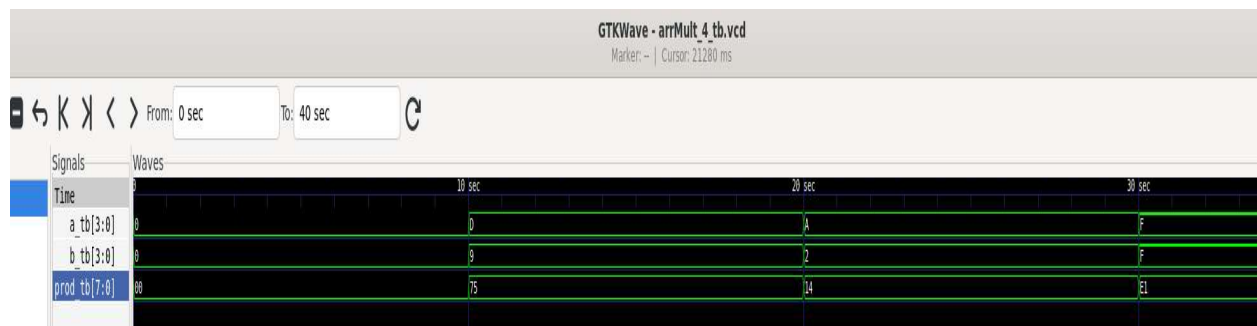
Project Introduction:

This project implements and designs a 4-bit array multiplier that multiplies two of 4-bits array and produce the answer of 8 bits. The array multiplier consists of half-adders and full-adders as shown below. Before using the full adder or half adder, we need a 2-to-1 mux selection before adding all the bits together as the rows. After connecting all of the modules, we need to verify that the circuit produces accurate results for all input combinations.

Design:

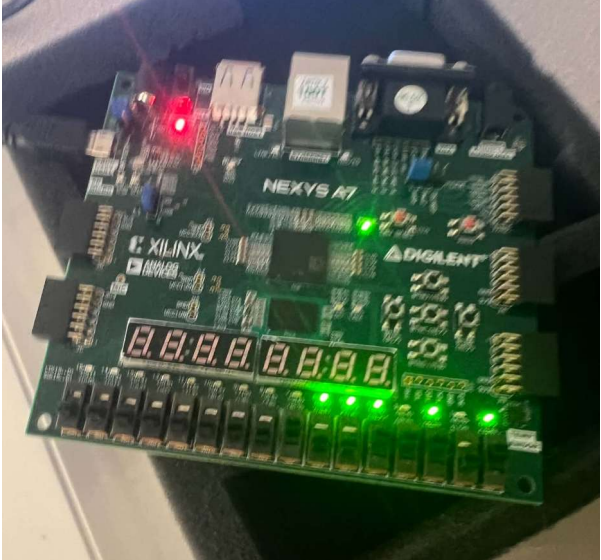
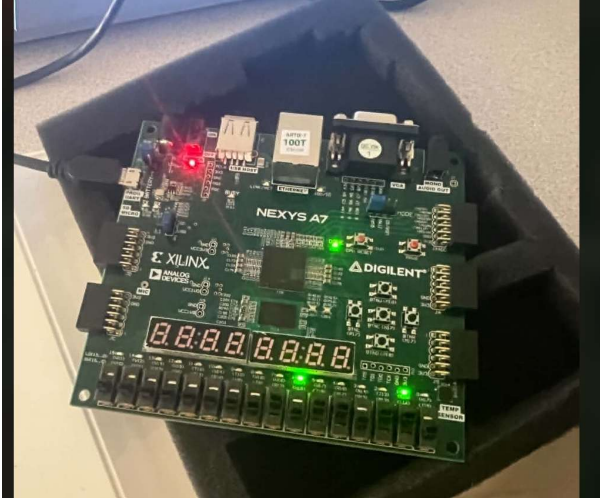
In this lab, I designed a 4-bit array multiplier that can multiply two 4-bit numbers and produce the result. The design is required to have a 2-to-1 mux, half adder, and full adder. The mux will select either the multiplication or zero, so basically like xor gate. If it is true (1), the output will be multiplication; otherwise, it will keep bitwise AND until it multiplies. Then the shift part is adding together using a half-adder and a full adder with the carry-out.

Waveform Results

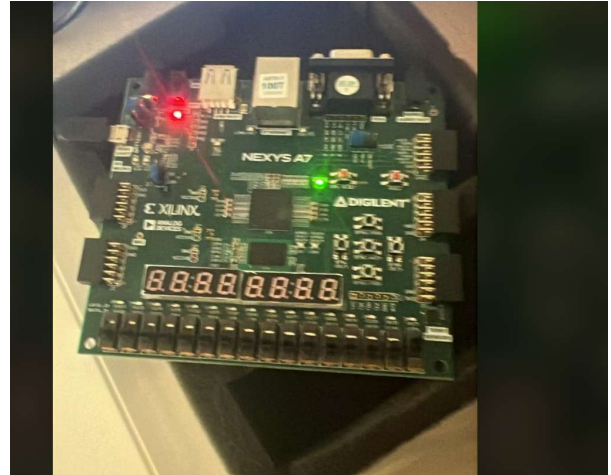


Multiplicand	Multiplier	Sum Output
(1101)b	(1001)b	(01110101)b
(0110)b	(1011)b	(01000010)b
(0001)b	(0000)b	(00000000)b
(1111)b	(0101)b	(01001011)b
(0100)b	(1010)b	(00101000)b
(1001)b	(0111)b	(00111111)b

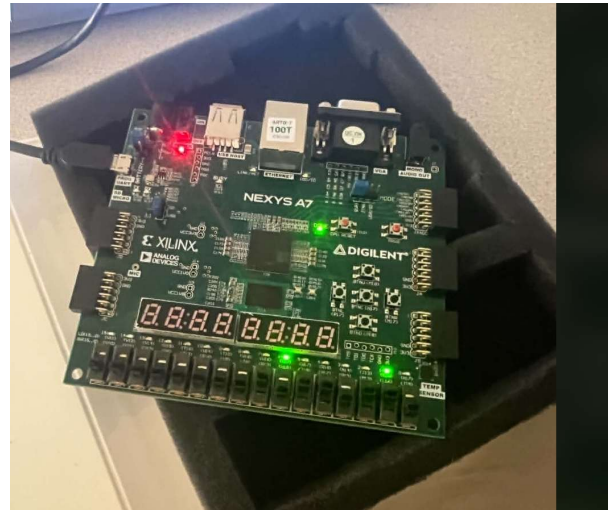
FPGA Board Implementation

Description	Picture
(1101)b time (1001)b	 A photograph of a green NEXYS A7 FPGA development board. The board features a 7-segment display at the bottom showing the time 11:01 in red. Above the display, there are several green LEDs and a red LED. The board is populated with various components, including a USB port, a power jack, and a small LCD screen. The text 'NEXYS A7' and 'XILINX' are visible on the board.
(0110)b time (1011)b	 A photograph of the same NEXYS A7 FPGA development board. The 7-segment display at the bottom now shows the time 10:11 in red. The board is connected to a power source, and the red LED is still lit. The text 'NEXYS A7' and 'XILINX' are visible on the board.

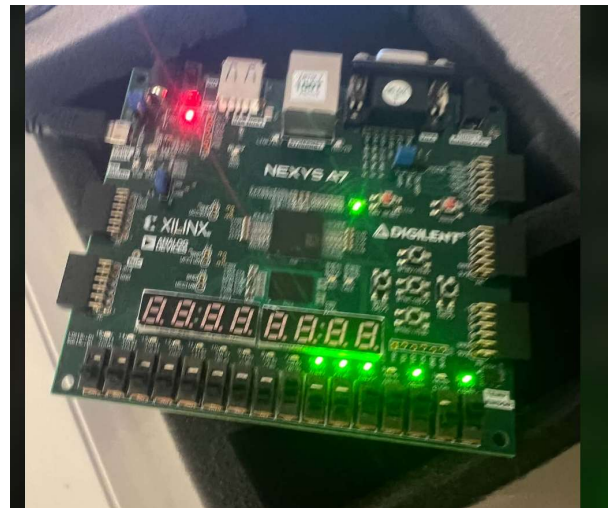
(0001)b time (0000)b



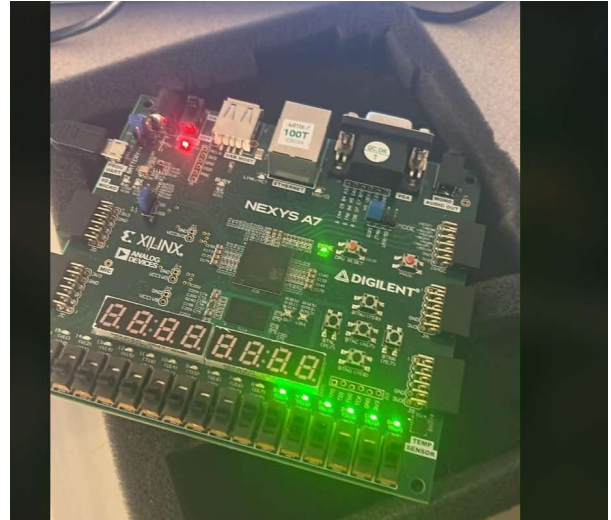
(1111)b time (0101)b



(0100)b time (1010)b



(1001)b time (0111)b



Conclusion

In the lab, I was successful in finishing all the project requirements and producing the correct. One of the problems of this project that I am facing is checking the correct result and verifying the answer. For some reason, all my results are wrong answer even though they have the same circuit. I figured out that I inserted a different row in one of the data sets, which led to the wrong answer. Overall, it is all doing fine for me, and I learned the concept behind it.

I/We certify that this submission is my/our original work."